

**Statistics 801**  
**Final Examination (100 points)**

Name \_\_\_\_\_

1. The price of a laptop of a particular make and configuration among dealers nationwide is assumed to have a Normal distribution with mean  $\mu = \$1200$  with standard deviation  $\sigma = \$50$ .
  - (a) (6) What is the probability that a laptop of the same make, chosen randomly from a dealer, will cost between \$1136 and \$1308 ? (Show your work).
  
  
  
  
  
  
  
  
  
  
  - (b) (6) What is the probability that the average price of 25 laptops of the same make, chosen randomly from a dealer, will be greater than \$1225? (Show your work).
  
  
  
  
  
  
  
  
  
  
2. To estimate the mean potency  $\mu$  of an antibiotic in a production batch, a manufacturer takes potency readings of 14 random samples, giving  $\bar{y} = 8.67$  , and  $s^2 = 0.56$ .
  - (a) (6) Estimate the mean potency for the batch using a 95% confidence interval.

- (b) (6) According to the industry standard, the standard deviation of the potency for this drug,  $\sigma$ , must not exceed 0.7. *State and test* the hypotheses that this batch does not meet the industry standard. Use  $\alpha = .05$ . Don't forget your conclusion and decision.

3. Brands of a type of electrical wire produced by two different manufacturers are being studied to compare their resistivity characteristics. The values of resistivity measurements obtained by the experimenter suggests that they can be considered to be samples from normal distributions and are used to calculate the following statistics:

| Brand A   |                        |                    | Brand B   |                        |                   |
|-----------|------------------------|--------------------|-----------|------------------------|-------------------|
| $n_1 = 8$ | $\bar{y}_{1.} = 0.140$ | $s_1^2 = 0.000007$ | $n_2 = 9$ | $\bar{y}_{2.} = 0.145$ | $s_2^2 = 0.00002$ |

Assuming that the population means are  $\mu_1$  and  $\mu_2$  and the population variances are  $\sigma_1^2$  and  $\sigma_2^2$ , respectively, answer the following questions:

- (a) (6) Test the hypothesis  $H_0 : \sigma_1^2 = \sigma_2^2$  vs.  $H_a : \sigma_1^2 \neq \sigma_2^2$ . Use  $\alpha = .05$  and show work.

- (b) (4) Now assume  $\sigma_1^2 = \sigma_2^2$ . Compute the standard error of the difference in the sample means  $\bar{y}_{1.} - \bar{y}_{2.}$ .

- (c) (6) Use the standard error from part (b) to construct a statistic to test the research hypothesis that Brand B wires have larger mean resistivity. State the null and alternative, the rejection region, and your decision using  $\alpha = .05$ . Show work.

- (d) (6) Use the standard error from part (b) to compute a 90% confidence interval for  $\mu_1 - \mu_2$ . Test the hypothesis in part (c) using this interval, stating the  $\alpha$ -level of the test.

4. An experiment was conducted to determine the relationship between the amount of warping in thin strips, ( $y$ ), made from a particular alloy and temperature, ( $x$ ) . A simple linear regression model of the form  $y = \beta_0 + \beta_1 x + \epsilon$  was first fitted to the data.

| Temperature<br>( $^{\circ}C$ ) ( $x$ ) | Amount of<br>Warping ( $y$ ) | $\bar{y}_i$ | $y_i - \bar{y}_i$ | $\sum(y_i - \bar{y}_i)^2$ | df |
|----------------------------------------|------------------------------|-------------|-------------------|---------------------------|----|
| 15                                     | 11, 11, 14                   | 12          |                   |                           |    |
| 20                                     | 14, 12, 13                   | 13          |                   |                           |    |
| 25                                     | 14, 12, 16                   | 14          |                   |                           |    |
| 30                                     | 18, 14, 16                   | 16          |                   |                           |    |
| 35                                     | 21, 19, 17                   | 19          |                   |                           |    |
| 40                                     | 24, 20, 22                   | 22          |                   |                           |    |
| 45                                     | 24, 23, 28                   | 25          |                   |                           |    |
| 50                                     | 31, 30, 32                   | 31          |                   |                           |    |

The following statistics have also been computed from the above data:

$$S_{xx} = 3150.0 \quad S_{yy} = 980.0 \quad S_{xy} = 1650.0$$

- (a) (10) Complete the analysis of variance table for the regression below. Use the **F-ratio** from the anova table to test  $H_0 : \beta_1 = 0$  vs.  $H_a : \beta_1 \neq 0$ , at  $\alpha = .05$ . Show work and state your conclusion.

| <u>Source</u> | <u>d.f.</u> | <u>SS</u> | <u>MS</u> | <u>F</u> |
|---------------|-------------|-----------|-----------|----------|
| Regression    |             |           |           |          |
| Error         |             |           |           |          |
| Total         | 23          |           |           |          |

- (b) (6) Compute the t-statistic to test the hypothesis that  $H_0 : \beta_1 = 0$  vs.  $H_a : \beta_1 \neq 0$ . Test this hypothesis using  $\alpha = .05$ . State the rejection region and your decision clearly. Show work.

- (c) (10) Perform the lack of fit test by completing the following table and state your conclusion. Use  $\alpha = .05$ .

| <u>Source</u> | <u>d.f.</u> | <u>SS</u> | <u>MS</u> | <u>F</u> |
|---------------|-------------|-----------|-----------|----------|
| Lack of Fit   |             |           |           |          |
| Pure Error    |             |           |           |          |
| Total Error   |             |           |           |          |

- (d) (3) Calculate the coefficient of determination.

(e) (3) Interpret the coefficient of determination in terms of the problem.

5. An agronomist wanted to see whether four chemical treatments were different in the number of soybean seeds that fail to emerge out of 100 seeds planted. She uniformly planted 100 soybean seeds treated with one of the four treatments in a randomly assigned plot in 5 different fields. She then counted the number of seeds that failed to emerge out of 100 seeds.

(a) (3) Write out the linear additive model for this study and define the terms. State the assumptions.

(b) (2) What type of experimental design is it? How do you know? Justify your answer.

(c) (7) Part of the ANOVA table is shown below. Fill in the missing values in the ANOVA table.

| Source             | degree of freedom | Sum of Squares | Mean of Squares | F-value | p-value |
|--------------------|-------------------|----------------|-----------------|---------|---------|
| Field              |                   |                | 21.925          |         | 0.0001  |
| Chemical Treatment |                   |                | 21.7833         |         | 0.0001  |
| Error              | 12                |                | 2.325           | -       | -       |
| Total              |                   |                | -               | -       | -       |

(d) (2) Based on the p-value and  $\alpha = 0.05$ , what can you conclude about the original research question?

(e) (2) Does it appear that blocking by field was effective? Support your answer.  $\alpha = 0.05$ .

6. What type of analyses (Contingency Table/Chi-square test or ANOVA or Regression) would you use for the following set of variables. Give an example for each.

(a) (2) Quantitative dependent, categorical independent

(b) (2) Categorical dependent, categorical independent

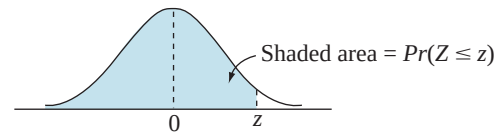
(c) (2) Quantitative dependent, quantitative independent

7. **BONUS QUESTIONS.** Each question is 1 point.

- (a) **(T/F)** In the analysis of covariance we compare the sum of squares of a full model with the sum of squares of a reduced model.
- (b) **(T/F)** The analysis of a factorial design is a better way to look at the interaction among treatments than the completely randomized design.
- (c) **(T/F)** The advantage of random forest over the linear regression is that the involving model is much simpler.
- (d) **(T/F)** The permutation test is different than the randomization test because in the permutation test we sample without replacement, and in the randomization test we sample with replacement.
- (e) **(T/F)** The Mann-Whitney test can only be used when the data are continuous.

**TABLE 1**

Standard normal curve areas



| <i>z</i> | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| −3.4     | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0003 | 0.0002 |
| −3.3     | 0.0005 | 0.0005 | 0.0005 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0004 | 0.0003 |
| −3.2     | 0.0007 | 0.0007 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0006 | 0.0005 | 0.0005 | 0.0005 |
| −3.1     | 0.0010 | 0.0009 | 0.0009 | 0.0009 | 0.0008 | 0.0008 | 0.0008 | 0.0008 | 0.0007 | 0.0007 |
| −3.0     | 0.0013 | 0.0013 | 0.0013 | 0.0012 | 0.0012 | 0.0011 | 0.0011 | 0.0011 | 0.0010 | 0.0010 |
| −2.9     | 0.0019 | 0.0018 | 0.0018 | 0.0017 | 0.0016 | 0.0016 | 0.0015 | 0.0015 | 0.0014 | 0.0014 |
| −2.8     | 0.0026 | 0.0025 | 0.0024 | 0.0023 | 0.0023 | 0.0022 | 0.0021 | 0.0021 | 0.0020 | 0.0019 |
| −2.7     | 0.0035 | 0.0034 | 0.0033 | 0.0032 | 0.0031 | 0.0030 | 0.0029 | 0.0028 | 0.0027 | 0.0026 |
| −2.6     | 0.0047 | 0.0045 | 0.0044 | 0.0043 | 0.0041 | 0.0040 | 0.0039 | 0.0038 | 0.0037 | 0.0036 |
| −2.5     | 0.0062 | 0.0060 | 0.0059 | 0.0057 | 0.0055 | 0.0054 | 0.0052 | 0.0051 | 0.0049 | 0.0048 |
| −2.4     | 0.0082 | 0.0080 | 0.0078 | 0.0075 | 0.0073 | 0.0071 | 0.0069 | 0.0068 | 0.0066 | 0.0064 |
| −2.3     | 0.0107 | 0.0104 | 0.0102 | 0.0099 | 0.0096 | 0.0094 | 0.0091 | 0.0089 | 0.0087 | 0.0084 |
| −2.2     | 0.0139 | 0.0136 | 0.0132 | 0.0129 | 0.0125 | 0.0122 | 0.0119 | 0.0116 | 0.0113 | 0.0110 |
| −2.1     | 0.0179 | 0.0174 | 0.0170 | 0.0166 | 0.0162 | 0.0158 | 0.0154 | 0.0150 | 0.0146 | 0.0143 |
| −2.0     | 0.0228 | 0.0222 | 0.0217 | 0.0212 | 0.0207 | 0.0202 | 0.0197 | 0.0192 | 0.0188 | 0.0183 |
| −1.9     | 0.0287 | 0.0281 | 0.0274 | 0.0268 | 0.0262 | 0.0256 | 0.0250 | 0.0244 | 0.0239 | 0.0233 |
| −1.8     | 0.0359 | 0.0351 | 0.0344 | 0.0336 | 0.0329 | 0.0322 | 0.0314 | 0.0307 | 0.0301 | 0.0294 |
| −1.7     | 0.0446 | 0.0436 | 0.0427 | 0.0418 | 0.0409 | 0.0401 | 0.0392 | 0.0384 | 0.0375 | 0.0367 |
| −1.6     | 0.0548 | 0.0537 | 0.0526 | 0.0516 | 0.0505 | 0.0495 | 0.0485 | 0.0475 | 0.0465 | 0.0455 |
| −1.5     | 0.0668 | 0.0655 | 0.0643 | 0.0630 | 0.0618 | 0.0606 | 0.0594 | 0.0582 | 0.0571 | 0.0559 |
| −1.4     | 0.0808 | 0.0793 | 0.0778 | 0.0764 | 0.0749 | 0.0735 | 0.0721 | 0.0708 | 0.0694 | 0.0681 |
| −1.3     | 0.0968 | 0.0951 | 0.0934 | 0.0918 | 0.0901 | 0.0885 | 0.0869 | 0.0853 | 0.0838 | 0.0823 |
| −1.2     | 0.1151 | 0.1131 | 0.1112 | 0.1093 | 0.1075 | 0.1056 | 0.1038 | 0.1020 | 0.1003 | 0.0985 |
| −1.1     | 0.1357 | 0.1335 | 0.1314 | 0.1292 | 0.1271 | 0.1251 | 0.1230 | 0.1210 | 0.1190 | 0.1170 |
| −1.0     | 0.1587 | 0.1562 | 0.1539 | 0.1515 | 0.1492 | 0.1469 | 0.1446 | 0.1423 | 0.1401 | 0.1379 |
| −0.9     | 0.1841 | 0.1814 | 0.1788 | 0.1762 | 0.1736 | 0.1711 | 0.1685 | 0.1660 | 0.1635 | 0.1611 |
| −0.8     | 0.2119 | 0.2090 | 0.2061 | 0.2033 | 0.2005 | 0.1977 | 0.1949 | 0.1922 | 0.1894 | 0.1867 |
| −0.7     | 0.2420 | 0.2389 | 0.2358 | 0.2327 | 0.2296 | 0.2266 | 0.2236 | 0.2206 | 0.2177 | 0.2148 |
| −0.6     | 0.2743 | 0.2709 | 0.2676 | 0.2643 | 0.2611 | 0.2578 | 0.2546 | 0.2514 | 0.2483 | 0.2451 |
| −0.5     | 0.3085 | 0.3050 | 0.3015 | 0.2981 | 0.2946 | 0.2912 | 0.2877 | 0.2843 | 0.2810 | 0.2776 |
| −0.4     | 0.3446 | 0.3409 | 0.3372 | 0.3336 | 0.3300 | 0.3264 | 0.3228 | 0.3192 | 0.3156 | 0.3121 |
| −0.3     | 0.3821 | 0.3783 | 0.3745 | 0.3707 | 0.3669 | 0.3632 | 0.3594 | 0.3557 | 0.3520 | 0.3483 |
| −0.2     | 0.4207 | 0.4168 | 0.4129 | 0.4090 | 0.4052 | 0.4013 | 0.3974 | 0.3936 | 0.3897 | 0.3859 |
| −0.1     | 0.4602 | 0.4562 | 0.4522 | 0.4483 | 0.4443 | 0.4404 | 0.4364 | 0.4325 | 0.4286 | 0.4247 |
| −0.0     | 0.5000 | 0.4960 | 0.4920 | 0.4880 | 0.4840 | 0.4801 | 0.4761 | 0.4721 | 0.4681 | 0.4641 |

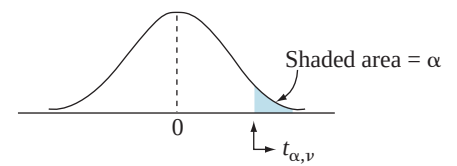
  

| <i>z</i> | Area       |
|----------|------------|
| −3.50    | 0.00023263 |
| −4.00    | 0.00003167 |
| −4.50    | 0.00000340 |
| −5.00    | 0.00000029 |

Source: Computed by M. Longnecker using Splus.

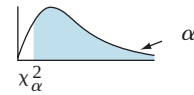




**TABLE 2**Percentage points of Student's  $t$  distribution

| $df/\alpha =$ | .40   | .25   | .10   | .05   | .025   | .01    | .005   | .001    | .0005   |
|---------------|-------|-------|-------|-------|--------|--------|--------|---------|---------|
| 1             | 0.325 | 1.000 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 | 318.309 | 636.619 |
| 2             | 0.289 | 0.816 | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  | 22.327  | 31.599  |
| 3             | 0.277 | 0.765 | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  | 10.215  | 12.924  |
| 4             | 0.271 | 0.741 | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |
| 5             | 0.267 | 0.727 | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |
| 6             | 0.265 | 0.718 | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |
| 7             | 0.263 | 0.711 | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |
| 8             | 0.262 | 0.706 | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |
| 9             | 0.261 | 0.703 | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |
| 10            | 0.260 | 0.700 | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |
| 11            | 0.260 | 0.697 | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |
| 12            | 0.259 | 0.695 | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |
| 13            | 0.259 | 0.694 | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |
| 14            | 0.258 | 0.692 | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |
| 15            | 0.258 | 0.691 | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |
| 16            | 0.258 | 0.690 | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |
| 17            | 0.257 | 0.689 | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |
| 18            | 0.257 | 0.688 | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |
| 19            | 0.257 | 0.688 | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |
| 20            | 0.257 | 0.687 | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |
| 21            | 0.257 | 0.686 | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |
| 22            | 0.256 | 0.686 | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |
| 23            | 0.256 | 0.685 | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  | 3.485   | 3.768   |
| 24            | 0.256 | 0.685 | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |
| 25            | 0.256 | 0.684 | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |
| 26            | 0.256 | 0.684 | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |
| 27            | 0.256 | 0.684 | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |
| 28            | 0.256 | 0.683 | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |
| 29            | 0.256 | 0.683 | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |
| 30            | 0.256 | 0.683 | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |
| 35            | 0.255 | 0.682 | 1.306 | 1.690 | 2.030  | 2.438  | 2.724  | 3.340   | 3.591   |
| 40            | 0.255 | 0.681 | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |
| 50            | 0.255 | 0.679 | 1.299 | 1.676 | 2.009  | 2.403  | 2.678  | 3.261   | 3.496   |
| 60            | 0.254 | 0.679 | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |
| 120           | 0.254 | 0.677 | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |
| inf.          | 0.253 | 0.674 | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |

Source: Computed by M. Longnecker using Splus.

**TABLE 7**

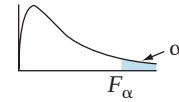
Percentage points of the chi-square distribution

| df  | $\alpha =$ | .999    | .995    | .99     | .975    | .95     | .90    |
|-----|------------|---------|---------|---------|---------|---------|--------|
| 1   |            | .000002 | .000039 | .000157 | .000982 | .003932 | .01579 |
| 2   |            | .002001 | .01003  | .02010  | .05064  | .1026   | .2107  |
| 3   |            | .02430  | .07172  | .1148   | .2158   | .3518   | .5844  |
| 4   |            | .09080  | .2070   | .2971   | .4844   | .7107   | 1.064  |
| 5   |            | .2102   | .4117   | .5543   | .8312   | 1.145   | 1.610  |
| 6   |            | .3811   | .6757   | .8721   | 1.237   | 1.635   | 2.204  |
| 7   |            | .5985   | .9893   | 1.239   | 1.690   | 2.167   | 2.833  |
| 8   |            | .8571   | 1.344   | 1.646   | 2.180   | 2.733   | 3.490  |
| 9   |            | 1.152   | 1.735   | 2.088   | 2.700   | 3.325   | 4.168  |
| 10  |            | 1.479   | 2.156   | 2.558   | 3.247   | 3.940   | 4.865  |
| 11  |            | 1.834   | 2.603   | 3.053   | 3.816   | 4.575   | 5.578  |
| 12  |            | 2.214   | 3.074   | 3.571   | 4.404   | 5.226   | 6.304  |
| 13  |            | 2.617   | 3.565   | 4.107   | 5.009   | 5.892   | 7.042  |
| 14  |            | 3.041   | 4.075   | 4.660   | 5.629   | 6.571   | 7.790  |
| 15  |            | 3.483   | 4.601   | 5.229   | 6.262   | 7.261   | 8.547  |
| 16  |            | 3.942   | 5.142   | 5.812   | 6.908   | 7.962   | 9.312  |
| 17  |            | 4.416   | 5.697   | 6.408   | 7.564   | 8.672   | 10.09  |
| 18  |            | 4.905   | 6.265   | 7.015   | 8.231   | 9.390   | 10.86  |
| 19  |            | 5.407   | 6.844   | 7.633   | 8.907   | 10.12   | 11.65  |
| 20  |            | 5.921   | 7.434   | 8.260   | 9.591   | 10.85   | 12.44  |
| 21  |            | 6.447   | 8.034   | 8.897   | 10.28   | 11.59   | 13.24  |
| 22  |            | 6.983   | 8.643   | 9.542   | 10.98   | 12.34   | 14.04  |
| 23  |            | 7.529   | 9.260   | 10.20   | 11.69   | 13.09   | 14.85  |
| 24  |            | 8.085   | 9.886   | 10.86   | 12.40   | 13.85   | 15.66  |
| 25  |            | 8.649   | 10.52   | 11.52   | 13.12   | 14.61   | 16.47  |
| 26  |            | 9.222   | 11.16   | 12.20   | 13.84   | 15.38   | 17.29  |
| 27  |            | 9.803   | 11.81   | 12.88   | 14.57   | 16.15   | 18.11  |
| 28  |            | 10.39   | 12.46   | 13.56   | 15.31   | 16.93   | 18.94  |
| 29  |            | 10.99   | 13.12   | 14.26   | 16.06   | 17.71   | 19.77  |
| 30  |            | 11.59   | 13.79   | 14.95   | 16.79   | 18.49   | 20.60  |
| 40  |            | 17.92   | 20.71   | 22.16   | 24.43   | 26.51   | 29.05  |
| 50  |            | 24.67   | 27.99   | 29.71   | 32.36   | 34.76   | 37.69  |
| 60  |            | 31.74   | 35.53   | 37.48   | 40.48   | 43.19   | 46.46  |
| 70  |            | 39.04   | 43.28   | 45.44   | 48.76   | 51.74   | 55.33  |
| 80  |            | 46.52   | 51.17   | 53.54   | 57.15   | 60.39   | 64.28  |
| 90  |            | 54.16   | 59.20   | 61.75   | 65.65   | 69.13   | 73.29  |
| 100 |            | 61.92   | 67.33   | 70.06   | 74.22   | 77.93   | 82.36  |
| 120 |            | 77.76   | 83.85   | 86.92   | 91.57   | 95.70   | 100.62 |
| 240 |            | 177.95  | 187.32  | 191.99  | 198.98  | 205.14  | 212.39 |

**TABLE 7**  
(continued)

| $\alpha = .10$ | .05    | .025   | .01    | .005   | .001   | df  |
|----------------|--------|--------|--------|--------|--------|-----|
| 2.706          | 3.841  | 5.024  | 6.635  | 7.879  | 10.83  | 1   |
| 4.605          | 5.991  | 7.378  | 9.210  | 10.60  | 13.82  | 2   |
| 6.251          | 7.815  | 9.348  | 11.34  | 12.84  | 16.27  | 3   |
| 7.779          | 9.488  | 11.14  | 13.28  | 14.86  | 18.47  | 4   |
| 9.236          | 11.07  | 12.83  | 15.09  | 16.75  | 20.52  | 5   |
| 10.64          | 12.59  | 14.45  | 16.81  | 18.55  | 22.46  | 6   |
| 12.02          | 14.07  | 16.01  | 18.48  | 20.28  | 24.32  | 7   |
| 13.36          | 15.51  | 17.53  | 20.09  | 21.95  | 26.12  | 8   |
| 14.68          | 16.92  | 19.02  | 21.67  | 23.59  | 27.88  | 9   |
| 15.99          | 18.31  | 20.48  | 23.21  | 25.19  | 29.59  | 10  |
| 17.28          | 19.68  | 21.92  | 24.72  | 26.76  | 31.27  | 11  |
| 18.55          | 21.03  | 23.34  | 26.22  | 28.30  | 32.91  | 12  |
| 19.81          | 22.36  | 24.74  | 27.69  | 29.82  | 34.53  | 13  |
| 21.06          | 23.68  | 26.12  | 29.14  | 31.32  | 36.12  | 14  |
| 22.31          | 25.00  | 27.49  | 30.58  | 32.80  | 37.70  | 15  |
| 23.54          | 26.30  | 28.85  | 32.00  | 34.27  | 39.25  | 16  |
| 24.77          | 27.59  | 30.19  | 33.41  | 35.72  | 40.79  | 17  |
| 25.99          | 28.87  | 31.53  | 34.81  | 37.16  | 42.31  | 18  |
| 27.20          | 30.14  | 32.85  | 36.19  | 38.58  | 43.82  | 19  |
| 28.41          | 31.41  | 34.17  | 37.57  | 40.00  | 45.31  | 20  |
| 29.62          | 32.67  | 35.48  | 38.93  | 41.40  | 46.80  | 21  |
| 30.81          | 33.92  | 36.78  | 40.29  | 42.80  | 48.27  | 22  |
| 32.01          | 35.17  | 38.08  | 41.64  | 44.18  | 49.73  | 23  |
| 33.20          | 36.42  | 39.36  | 42.98  | 45.56  | 51.18  | 24  |
| 34.38          | 37.65  | 40.65  | 44.31  | 46.93  | 52.62  | 25  |
| 35.56          | 38.89  | 41.92  | 45.64  | 48.29  | 54.05  | 26  |
| 36.74          | 40.11  | 43.19  | 46.96  | 49.65  | 55.48  | 27  |
| 37.92          | 41.34  | 44.46  | 48.28  | 50.99  | 56.89  | 28  |
| 39.09          | 42.56  | 45.72  | 49.59  | 52.34  | 58.30  | 29  |
| 40.26          | 43.77  | 46.98  | 50.89  | 53.67  | 59.70  | 30  |
| 51.81          | 55.76  | 59.34  | 63.69  | 66.77  | 73.40  | 40  |
| 63.17          | 67.50  | 71.42  | 76.15  | 79.49  | 86.66  | 50  |
| 74.40          | 79.08  | 83.30  | 88.38  | 91.95  | 99.61  | 60  |
| 85.53          | 90.53  | 95.02  | 100.43 | 104.21 | 112.32 | 70  |
| 96.58          | 101.88 | 106.63 | 112.33 | 116.32 | 124.84 | 80  |
| 107.57         | 113.15 | 118.14 | 124.12 | 128.30 | 137.21 | 90  |
| 118.50         | 124.34 | 129.56 | 135.81 | 140.17 | 149.45 | 100 |
| 140.23         | 146.57 | 152.21 | 158.95 | 163.65 | 173.62 | 120 |
| 268.47         | 277.14 | 284.80 | 293.89 | 300.18 | 313.44 | 240 |

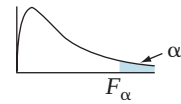
Source: Computed by P. J. Hildebrand.

**TABLE 8**Percentage points of the  $F$  distribution ( $df_2$  between 7 and 12)

| $df_2$    | $\alpha$ | $df_1$ |       |       |       |       |       |       |       |       |       |
|-----------|----------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|           |          | 1      | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |
| <b>7</b>  | .25      | 1.57   | 1.70  | 1.72  | 1.72  | 1.71  | 1.71  | 1.70  | 1.70  | 1.69  | 1.69  |
|           | .10      | 3.59   | 3.26  | 3.07  | 2.96  | 2.88  | 2.83  | 2.78  | 2.75  | 2.72  | 2.70  |
|           | .05      | 5.59   | 4.74  | 4.35  | 4.12  | 3.97  | 3.87  | 3.79  | 3.73  | 3.68  | 3.64  |
|           | .025     | 8.07   | 6.54  | 5.89  | 5.52  | 5.29  | 5.12  | 4.99  | 4.90  | 4.82  | 4.76  |
|           | .01      | 12.25  | 9.55  | 8.45  | 7.85  | 7.46  | 7.19  | 6.99  | 6.84  | 6.72  | 6.62  |
|           | .005     | 16.24  | 12.40 | 10.88 | 10.05 | 9.52  | 9.16  | 8.89  | 8.68  | 8.51  | 8.38  |
|           | .001     | 29.25  | 21.69 | 18.77 | 17.20 | 16.21 | 15.52 | 15.02 | 14.63 | 14.33 | 14.08 |
| <b>8</b>  | .25      | 1.54   | 1.66  | 1.67  | 1.66  | 1.66  | 1.65  | 1.64  | 1.64  | 1.63  | 1.63  |
|           | .10      | 3.46   | 3.11  | 2.92  | 2.81  | 2.73  | 2.67  | 2.62  | 2.59  | 2.56  | 2.54  |
|           | .05      | 5.32   | 4.46  | 4.07  | 3.84  | 3.69  | 3.58  | 3.50  | 3.44  | 3.39  | 3.35  |
|           | .025     | 7.57   | 6.06  | 5.42  | 5.05  | 4.82  | 4.65  | 4.53  | 4.43  | 4.36  | 4.30  |
|           | .01      | 11.26  | 8.65  | 7.59  | 7.01  | 6.63  | 6.37  | 6.18  | 6.03  | 5.91  | 5.81  |
|           | .005     | 14.69  | 11.04 | 9.60  | 8.81  | 8.30  | 7.95  | 7.69  | 7.50  | 7.34  | 7.21  |
|           | .001     | 25.41  | 18.49 | 15.83 | 14.39 | 13.48 | 12.86 | 12.40 | 12.05 | 11.77 | 11.54 |
| <b>9</b>  | .25      | 1.51   | 1.62  | 1.63  | 1.63  | 1.62  | 1.61  | 1.60  | 1.60  | 1.59  | 1.59  |
|           | .10      | 3.36   | 3.01  | 2.81  | 2.69  | 2.61  | 2.55  | 2.51  | 2.47  | 2.44  | 2.42  |
|           | .05      | 5.12   | 4.26  | 3.86  | 3.63  | 3.48  | 3.37  | 3.29  | 3.23  | 3.18  | 3.14  |
|           | .025     | 7.21   | 5.71  | 5.08  | 4.72  | 4.48  | 4.32  | 4.20  | 4.10  | 4.03  | 3.96  |
|           | .01      | 10.56  | 8.02  | 6.99  | 6.42  | 6.06  | 5.80  | 5.61  | 5.47  | 5.35  | 5.26  |
|           | .005     | 13.61  | 10.11 | 8.72  | 7.96  | 7.47  | 7.13  | 6.88  | 6.69  | 6.54  | 6.42  |
|           | .001     | 22.86  | 16.39 | 13.90 | 12.56 | 11.71 | 11.13 | 10.70 | 10.37 | 10.11 | 9.89  |
| <b>10</b> | .25      | 1.49   | 1.60  | 1.60  | 1.59  | 1.59  | 1.58  | 1.57  | 1.56  | 1.56  | 1.55  |
|           | .10      | 3.29   | 2.92  | 2.73  | 2.61  | 2.52  | 2.46  | 2.41  | 2.38  | 2.35  | 2.32  |
|           | .05      | 4.96   | 4.10  | 3.71  | 3.48  | 3.33  | 3.22  | 3.14  | 3.07  | 3.02  | 2.98  |
|           | .025     | 6.94   | 5.46  | 4.83  | 4.47  | 4.24  | 4.07  | 3.95  | 3.85  | 3.78  | 3.72  |
|           | .01      | 10.04  | 7.56  | 6.55  | 5.99  | 5.64  | 5.39  | 5.20  | 5.06  | 4.94  | 4.85  |
|           | .005     | 12.83  | 9.43  | 8.08  | 7.34  | 6.87  | 6.54  | 6.30  | 6.12  | 5.97  | 5.85  |
|           | .001     | 21.04  | 14.91 | 12.55 | 11.28 | 10.48 | 9.93  | 9.52  | 9.20  | 8.96  | 8.75  |
| <b>11</b> | .25      | 1.47   | 1.58  | 1.58  | 1.57  | 1.56  | 1.55  | 1.54  | 1.53  | 1.53  | 1.52  |
|           | .10      | 3.23   | 2.86  | 2.66  | 2.54  | 2.45  | 2.39  | 2.34  | 2.30  | 2.27  | 2.25  |
|           | .05      | 4.84   | 3.98  | 3.59  | 3.36  | 3.20  | 3.09  | 3.01  | 2.95  | 2.90  | 2.85  |
|           | .025     | 6.72   | 5.26  | 4.63  | 4.28  | 4.04  | 3.88  | 3.76  | 3.66  | 3.59  | 3.53  |
|           | .01      | 9.65   | 7.21  | 6.22  | 5.67  | 5.32  | 5.07  | 4.89  | 4.74  | 4.63  | 4.54  |
|           | .005     | 12.23  | 8.91  | 7.60  | 6.88  | 6.42  | 6.10  | 5.86  | 5.68  | 5.54  | 5.42  |
|           | .001     | 19.69  | 13.81 | 11.56 | 10.35 | 9.58  | 9.05  | 8.66  | 8.35  | 8.12  | 7.92  |
| <b>12</b> | .25      | 1.46   | 1.56  | 1.56  | 1.55  | 1.54  | 1.53  | 1.52  | 1.51  | 1.51  | 1.50  |
|           | .10      | 3.18   | 2.81  | 2.61  | 2.48  | 2.39  | 2.33  | 2.28  | 2.24  | 2.21  | 2.19  |
|           | .05      | 4.75   | 3.89  | 3.49  | 3.26  | 3.11  | 3.00  | 2.91  | 2.85  | 2.80  | 2.75  |
|           | .025     | 6.55   | 5.10  | 4.47  | 4.12  | 3.89  | 3.73  | 3.61  | 3.51  | 3.44  | 3.37  |
|           | .01      | 9.33   | 6.93  | 5.95  | 5.41  | 5.06  | 4.82  | 4.64  | 4.50  | 4.39  | 4.30  |
|           | .005     | 11.75  | 8.51  | 7.23  | 6.52  | 6.07  | 5.76  | 5.52  | 5.35  | 5.20  | 5.09  |
|           | .001     | 18.64  | 12.97 | 10.80 | 9.63  | 8.89  | 8.38  | 8.00  | 7.71  | 7.48  | 7.29  |

**TABLE 8**  
(continued)

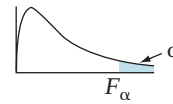
| df <sub>1</sub> |       |       |       |       |       |       |       |       |       |          | df <sub>2</sub> |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|-----------------|
| 12              | 15    | 20    | 24    | 30    | 40    | 60    | 120   | 240   | inf.  | $\alpha$ |                 |
| 1.68            | 1.68  | 1.67  | 1.67  | 1.66  | 1.66  | 1.65  | 1.65  | 1.65  | 1.65  | .25      | <b>7</b>        |
| 2.67            | 2.63  | 2.59  | 2.58  | 2.56  | 2.54  | 2.51  | 2.49  | 2.48  | 2.47  | .10      |                 |
| 3.57            | 3.51  | 3.44  | 3.41  | 3.38  | 3.34  | 3.30  | 3.27  | 3.25  | 3.23  | .05      |                 |
| 4.67            | 4.57  | 4.47  | 4.41  | 4.36  | 4.31  | 4.25  | 4.20  | 4.17  | 4.14  | .025     |                 |
| 6.47            | 6.31  | 6.16  | 6.07  | 5.99  | 5.91  | 5.82  | 5.74  | 5.69  | 5.65  | .01      |                 |
| 8.18            | 7.97  | 7.75  | 7.64  | 7.53  | 7.42  | 7.31  | 7.19  | 7.13  | 7.08  | .005     |                 |
| 13.71           | 13.32 | 12.93 | 12.73 | 12.53 | 12.33 | 12.12 | 11.91 | 11.80 | 11.70 | .001     |                 |
| 1.62            | 1.62  | 1.61  | 1.60  | 1.60  | 1.59  | 1.59  | 1.58  | 1.58  | 1.58  | .25      | <b>8</b>        |
| 2.50            | 2.46  | 2.42  | 2.40  | 2.38  | 2.36  | 2.34  | 2.32  | 2.30  | 2.29  | .10      |                 |
| 3.28            | 3.22  | 3.15  | 3.12  | 3.08  | 3.04  | 3.01  | 2.97  | 2.95  | 2.93  | .05      |                 |
| 4.20            | 4.10  | 4.00  | 3.95  | 3.89  | 3.84  | 3.78  | 3.73  | 3.70  | 3.67  | .025     |                 |
| 5.67            | 5.52  | 5.36  | 5.28  | 5.20  | 5.12  | 5.03  | 4.95  | 4.90  | 4.86  | .01      |                 |
| 7.01            | 6.81  | 6.61  | 6.50  | 6.40  | 6.29  | 6.18  | 6.06  | 6.01  | 5.95  | .005     |                 |
| 11.19           | 10.84 | 10.48 | 10.30 | 10.11 | 9.92  | 9.73  | 9.53  | 9.43  | 9.33  | .001     |                 |
| 1.58            | 1.57  | 1.56  | 1.56  | 1.55  | 1.54  | 1.64  | 1.53  | 1.53  | 1.53  | .25      | <b>9</b>        |
| 2.38            | 2.34  | 2.30  | 2.28  | 2.25  | 2.23  | 2.21  | 2.18  | 2.17  | 2.16  | .10      |                 |
| 3.07            | 3.01  | 2.94  | 2.90  | 2.86  | 2.83  | 2.79  | 2.75  | 2.73  | 2.71  | .05      |                 |
| 3.87            | 3.77  | 3.67  | 3.61  | 3.56  | 3.51  | 3.45  | 3.39  | 3.36  | 3.33  | .025     |                 |
| 5.11            | 4.96  | 4.81  | 4.73  | 4.65  | 4.57  | 4.48  | 4.40  | 4.35  | 4.31  | .01      |                 |
| 6.23            | 6.03  | 5.83  | 5.73  | 5.62  | 5.52  | 5.41  | 5.30  | 5.24  | 5.19  | .005     |                 |
| 9.57            | 9.24  | 8.90  | 8.72  | 8.55  | 8.37  | 8.19  | 8.00  | 7.91  | 7.81  | .001     |                 |
| 1.54            | 1.53  | 1.52  | 1.52  | 1.51  | 1.51  | 1.50  | 1.49  | 1.49  | 1.48  | .25      | <b>10</b>       |
| 2.28            | 2.24  | 2.20  | 2.18  | 2.16  | 2.13  | 2.11  | 2.08  | 2.07  | 2.06  | .10      |                 |
| 2.91            | 2.85  | 2.77  | 2.74  | 2.70  | 2.66  | 2.62  | 2.58  | 2.56  | 2.54  | .05      |                 |
| 3.62            | 3.52  | 3.42  | 3.37  | 3.31  | 3.26  | 3.20  | 3.14  | 3.11  | 3.08  | .025     |                 |
| 4.71            | 4.56  | 4.41  | 4.33  | 4.25  | 4.17  | 4.08  | 4.00  | 3.95  | 3.91  | .01      |                 |
| 5.66            | 5.47  | 5.27  | 5.17  | 5.07  | 4.97  | 4.86  | 4.75  | 4.69  | 4.64  | .005     |                 |
| 8.45            | 8.13  | 7.80  | 7.64  | 7.47  | 7.30  | 7.12  | 6.94  | 6.85  | 6.76  | .001     |                 |
| 1.51            | 1.50  | 1.49  | 1.49  | 1.48  | 1.47  | 1.47  | 1.46  | 1.45  | 1.45  | .25      | <b>11</b>       |
| 2.21            | 2.17  | 2.12  | 2.10  | 2.08  | 2.05  | 2.03  | 2.00  | 1.99  | 1.97  | .10      |                 |
| 2.79            | 2.72  | 2.65  | 2.61  | 2.57  | 2.53  | 2.49  | 2.45  | 2.43  | 2.40  | .05      |                 |
| 3.43            | 3.33  | 3.23  | 3.17  | 3.12  | 3.06  | 3.00  | 2.94  | 2.91  | 2.88  | .025     |                 |
| 4.40            | 4.25  | 4.10  | 4.02  | 3.94  | 3.86  | 3.78  | 3.69  | 3.65  | 3.60  | .01      |                 |
| 5.24            | 5.05  | 4.86  | 4.76  | 4.65  | 4.55  | 4.45  | 4.34  | 4.28  | 4.23  | .005     |                 |
| 7.63            | 7.32  | 7.01  | 6.85  | 6.68  | 6.52  | 6.35  | 6.18  | 6.09  | 6.00  | .001     |                 |
| 1.49            | 1.48  | 1.47  | 1.46  | 1.45  | 1.45  | 1.44  | 1.43  | 1.43  | 1.42  | .25      | <b>12</b>       |
| 2.15            | 2.10  | 2.06  | 2.04  | 2.01  | 1.99  | 1.96  | 1.93  | 1.92  | 1.90  | .10      |                 |
| 2.69            | 2.62  | 2.54  | 2.51  | 2.47  | 2.43  | 2.38  | 2.34  | 2.32  | 2.30  | .05      |                 |
| 3.28            | 3.18  | 3.07  | 3.02  | 2.96  | 2.91  | 2.85  | 2.79  | 2.76  | 2.72  | .025     |                 |
| 4.16            | 4.01  | 3.86  | 3.78  | 3.70  | 3.62  | 3.54  | 3.45  | 3.41  | 3.36  | .01      |                 |
| 4.91            | 4.72  | 4.53  | 4.43  | 4.33  | 4.23  | 4.12  | 4.01  | 3.96  | 3.90  | .005     |                 |
| 7.00            | 6.71  | 6.40  | 6.25  | 6.09  | 5.93  | 5.76  | 5.59  | 5.51  | 5.42  | .001     |                 |

**TABLE 8**Percentage points of the  $F$  distribution ( $df_2$  between 13 and 18)

| $df_2$    | $\alpha$ | $df_1$ |       |       |      |      |      |      |      |      |      |
|-----------|----------|--------|-------|-------|------|------|------|------|------|------|------|
|           |          | 1      | 2     | 3     | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
| <b>13</b> | .25      | 1.45   | 1.55  | 1.55  | 1.53 | 1.52 | 1.51 | 1.50 | 1.49 | 1.49 | 1.48 |
|           | .10      | 3.14   | 2.76  | 2.56  | 2.43 | 2.35 | 2.28 | 2.23 | 2.20 | 2.16 | 2.14 |
|           | .05      | 4.67   | 3.81  | 3.41  | 3.18 | 3.03 | 2.92 | 2.83 | 2.77 | 2.71 | 2.67 |
|           | .025     | 6.41   | 4.97  | 4.35  | 4.00 | 3.77 | 3.60 | 3.48 | 3.39 | 3.31 | 3.25 |
|           | .01      | 9.07   | 6.70  | 5.74  | 5.21 | 4.86 | 4.62 | 4.44 | 4.30 | 4.19 | 4.10 |
|           | .005     | 11.37  | 8.19  | 6.93  | 6.23 | 5.79 | 5.48 | 5.25 | 5.08 | 4.94 | 4.82 |
|           | .001     | 17.82  | 12.31 | 10.21 | 9.07 | 8.35 | 7.86 | 7.49 | 7.21 | 6.98 | 6.80 |
| <b>14</b> | .25      | 1.44   | 1.53  | 1.53  | 1.52 | 1.51 | 1.50 | 1.49 | 1.48 | 1.47 | 1.46 |
|           | .10      | 3.10   | 2.73  | 2.52  | 2.39 | 2.31 | 2.24 | 2.19 | 2.15 | 2.12 | 2.10 |
|           | .05      | 4.60   | 3.74  | 3.34  | 3.11 | 2.96 | 2.85 | 2.76 | 2.70 | 2.65 | 2.60 |
|           | .025     | 6.30   | 4.86  | 4.24  | 3.89 | 3.66 | 3.50 | 3.38 | 3.29 | 3.21 | 3.15 |
|           | .01      | 8.86   | 6.51  | 5.56  | 5.04 | 4.69 | 4.46 | 4.28 | 4.14 | 4.03 | 3.94 |
|           | .005     | 11.06  | 7.92  | 6.68  | 6.00 | 5.56 | 5.26 | 5.03 | 4.86 | 4.72 | 4.60 |
|           | .001     | 17.14  | 11.78 | 9.73  | 8.62 | 7.92 | 7.44 | 7.08 | 6.80 | 6.58 | 6.40 |
| <b>15</b> | .25      | 1.43   | 1.52  | 1.52  | 1.51 | 1.49 | 1.48 | 1.47 | 1.46 | 1.46 | 1.45 |
|           | .10      | 3.07   | 2.70  | 2.49  | 2.36 | 2.27 | 2.21 | 2.16 | 2.12 | 2.09 | 2.06 |
|           | .05      | 4.54   | 3.68  | 3.29  | 3.06 | 2.90 | 2.79 | 2.71 | 2.64 | 2.59 | 2.54 |
|           | .025     | 6.20   | 4.77  | 4.15  | 3.80 | 3.58 | 3.41 | 3.29 | 3.20 | 3.12 | 3.06 |
|           | .01      | 8.68   | 6.36  | 5.42  | 4.89 | 4.56 | 4.32 | 4.14 | 4.00 | 3.89 | 3.80 |
|           | .005     | 10.80  | 7.70  | 6.48  | 5.80 | 5.37 | 5.07 | 4.85 | 4.67 | 4.54 | 4.42 |
|           | .001     | 16.59  | 11.34 | 9.34  | 8.25 | 7.57 | 7.09 | 6.74 | 6.47 | 6.26 | 6.08 |
| <b>16</b> | .25      | 1.42   | 1.51  | 1.51  | 1.50 | 1.48 | 1.47 | 1.46 | 1.45 | 1.44 | 1.44 |
|           | .10      | 3.05   | 2.67  | 2.46  | 2.33 | 2.24 | 2.18 | 2.13 | 2.09 | 2.06 | 2.03 |
|           | .05      | 4.49   | 3.63  | 3.24  | 3.01 | 2.85 | 2.74 | 2.66 | 2.59 | 2.54 | 2.49 |
|           | .025     | 6.12   | 4.69  | 4.08  | 3.73 | 3.50 | 3.34 | 3.22 | 3.12 | 3.05 | 2.99 |
|           | .01      | 8.53   | 6.23  | 5.29  | 4.77 | 4.44 | 4.20 | 4.03 | 3.89 | 3.78 | 3.69 |
|           | .005     | 10.58  | 7.51  | 6.30  | 5.64 | 5.21 | 4.91 | 4.69 | 4.52 | 4.38 | 4.27 |
|           | .001     | 16.12  | 10.97 | 9.01  | 7.94 | 7.27 | 6.80 | 6.46 | 6.19 | 5.98 | 5.81 |
| <b>17</b> | .25      | 1.42   | 1.51  | 1.50  | 1.49 | 1.47 | 1.46 | 1.45 | 1.44 | 1.43 | 1.43 |
|           | .10      | 3.03   | 2.64  | 2.44  | 2.31 | 2.22 | 2.15 | 2.10 | 2.06 | 2.03 | 2.00 |
|           | .05      | 4.45   | 3.59  | 3.20  | 2.96 | 2.81 | 2.70 | 2.61 | 2.55 | 2.49 | 2.45 |
|           | .025     | 6.04   | 4.62  | 4.01  | 3.66 | 3.44 | 3.28 | 3.16 | 3.06 | 2.98 | 2.92 |
|           | .01      | 8.40   | 6.11  | 5.18  | 4.67 | 4.34 | 4.10 | 3.93 | 3.79 | 3.68 | 3.59 |
|           | .005     | 10.38  | 7.35  | 6.16  | 5.50 | 5.07 | 4.78 | 4.56 | 4.39 | 4.25 | 4.14 |
|           | .001     | 15.72  | 10.66 | 8.73  | 7.68 | 7.02 | 6.56 | 6.22 | 5.96 | 5.75 | 5.58 |
| <b>18</b> | .25      | 1.41   | 1.50  | 1.49  | 1.48 | 1.46 | 1.45 | 1.44 | 1.43 | 1.42 | 1.42 |
|           | .10      | 3.01   | 2.62  | 2.42  | 2.29 | 2.20 | 2.13 | 2.08 | 2.04 | 2.00 | 1.98 |
|           | .05      | 4.41   | 3.55  | 3.16  | 2.93 | 2.77 | 2.66 | 2.58 | 2.51 | 2.46 | 2.41 |
|           | .025     | 5.98   | 4.56  | 3.95  | 3.61 | 3.38 | 3.22 | 3.10 | 3.01 | 2.93 | 2.87 |
|           | .01      | 8.29   | 6.01  | 5.09  | 4.58 | 4.25 | 4.01 | 3.84 | 3.71 | 3.60 | 3.51 |
|           | .005     | 10.22  | 7.21  | 6.03  | 5.37 | 4.96 | 4.66 | 4.44 | 4.28 | 4.14 | 4.03 |
|           | .001     | 15.38  | 10.39 | 8.49  | 7.46 | 6.81 | 6.35 | 6.02 | 5.76 | 5.56 | 5.39 |

**TABLE 8**  
(continued)

| df <sub>1</sub> |      |      |      |      |      |      |      |      |      |          | df <sub>2</sub> |
|-----------------|------|------|------|------|------|------|------|------|------|----------|-----------------|
| 12              | 15   | 20   | 24   | 30   | 40   | 60   | 120  | 240  | inf. | $\alpha$ |                 |
| 1.47            | 1.46 | 1.45 | 1.44 | 1.43 | 1.42 | 1.42 | 1.41 | 1.40 | 1.40 | .25      | <b>13</b>       |
| 2.10            | 2.05 | 2.01 | 1.98 | 1.96 | 1.93 | 1.90 | 1.88 | 1.86 | 1.85 | .10      |                 |
| 2.60            | 2.53 | 2.46 | 2.42 | 2.38 | 2.34 | 2.30 | 2.25 | 2.23 | 2.21 | .05      |                 |
| 3.15            | 3.05 | 2.95 | 2.89 | 2.84 | 2.78 | 2.72 | 2.66 | 2.63 | 2.60 | .025     |                 |
| 3.96            | 3.82 | 3.66 | 3.59 | 3.51 | 3.43 | 3.34 | 3.25 | 3.21 | 3.17 | .01      |                 |
| 4.64            | 4.46 | 4.27 | 4.17 | 4.07 | 3.97 | 3.87 | 3.76 | 3.70 | 3.65 | .005     |                 |
| 6.52            | 6.23 | 5.93 | 5.78 | 5.63 | 5.47 | 5.30 | 5.14 | 5.05 | 4.97 | .001     |                 |
| 1.45            | 1.44 | 1.43 | 1.42 | 1.41 | 1.41 | 1.40 | 1.39 | 1.38 | 1.38 | .25      | <b>14</b>       |
| 2.05            | 2.01 | 1.96 | 1.94 | 1.91 | 1.89 | 1.86 | 1.83 | 1.81 | 1.80 | .10      |                 |
| 2.53            | 2.46 | 2.39 | 2.35 | 2.31 | 2.27 | 2.22 | 2.18 | 2.15 | 2.13 | .05      |                 |
| 3.05            | 2.95 | 2.84 | 2.79 | 2.73 | 2.67 | 2.61 | 2.55 | 2.52 | 2.49 | .025     |                 |
| 3.80            | 3.66 | 3.51 | 3.43 | 3.35 | 3.27 | 3.18 | 3.09 | 3.05 | 3.00 | .01      |                 |
| 4.43            | 4.25 | 4.06 | 3.96 | 3.86 | 3.76 | 3.66 | 3.55 | 3.49 | 3.44 | .005     |                 |
| 6.13            | 5.85 | 5.56 | 5.41 | 5.25 | 5.10 | 4.94 | 4.77 | 4.69 | 4.60 | .001     |                 |
| 1.44            | 1.43 | 1.41 | 1.41 | 1.40 | 1.39 | 1.38 | 1.37 | 1.36 | 1.36 | .25      | <b>15</b>       |
| 2.02            | 1.97 | 1.92 | 1.90 | 1.87 | 1.85 | 1.82 | 1.79 | 1.77 | 1.76 | .10      |                 |
| 2.48            | 2.40 | 2.33 | 2.29 | 2.25 | 2.20 | 2.16 | 2.11 | 2.09 | 2.07 | .05      |                 |
| 2.96            | 2.86 | 2.76 | 2.70 | 2.64 | 2.59 | 2.52 | 2.46 | 2.43 | 2.40 | .025     |                 |
| 3.67            | 3.52 | 3.37 | 3.29 | 3.21 | 3.13 | 3.05 | 2.96 | 2.91 | 2.87 | .01      |                 |
| 4.25            | 4.07 | 3.88 | 3.79 | 3.69 | 3.58 | 3.48 | 3.37 | 3.32 | 3.26 | .005     |                 |
| 5.81            | 5.54 | 5.25 | 5.10 | 4.95 | 4.80 | 4.64 | 4.47 | 4.39 | 4.31 | .001     |                 |
| 1.43            | 1.41 | 1.40 | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.35 | 1.34 | .25      | <b>16</b>       |
| 1.99            | 1.94 | 1.89 | 1.87 | 1.84 | 1.81 | 1.78 | 1.75 | 1.73 | 1.72 | .10      |                 |
| 2.42            | 2.35 | 2.28 | 2.24 | 2.19 | 2.15 | 2.11 | 2.06 | 2.03 | 2.01 | .05      |                 |
| 2.89            | 2.79 | 2.68 | 2.63 | 2.57 | 2.51 | 2.45 | 2.38 | 2.35 | 2.32 | .025     |                 |
| 3.55            | 3.41 | 3.26 | 3.18 | 3.10 | 3.02 | 2.93 | 2.84 | 2.80 | 2.75 | .01      |                 |
| 4.10            | 3.92 | 3.73 | 3.64 | 3.54 | 3.44 | 3.33 | 3.22 | 3.17 | 3.11 | .005     |                 |
| 5.55            | 5.27 | 4.99 | 4.85 | 4.70 | 4.54 | 4.39 | 4.23 | 4.14 | 4.06 | .001     |                 |
| 1.41            | 1.40 | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.34 | 1.33 | 1.33 | .25      | <b>17</b>       |
| 1.96            | 1.91 | 1.86 | 1.84 | 1.81 | 1.78 | 1.75 | 1.72 | 1.70 | 1.69 | .10      |                 |
| 2.38            | 2.31 | 2.23 | 2.19 | 2.15 | 2.10 | 2.06 | 2.01 | 1.99 | 1.96 | .05      |                 |
| 2.82            | 2.72 | 2.62 | 2.56 | 2.50 | 2.44 | 2.38 | 2.32 | 2.28 | 2.25 | .025     |                 |
| 3.46            | 3.31 | 3.16 | 3.08 | 3.00 | 2.92 | 2.83 | 2.75 | 2.70 | 2.65 | .01      |                 |
| 3.97            | 3.79 | 3.61 | 3.51 | 3.41 | 3.31 | 3.21 | 3.10 | 3.04 | 2.98 | .005     |                 |
| 5.32            | 5.05 | 4.78 | 4.63 | 4.48 | 4.33 | 4.18 | 4.02 | 3.93 | 3.85 | .001     |                 |
| 1.40            | 1.39 | 1.38 | 1.37 | 1.36 | 1.35 | 1.34 | 1.33 | 1.32 | 1.32 | .25      | <b>18</b>       |
| 1.93            | 1.89 | 1.84 | 1.81 | 1.78 | 1.75 | 1.72 | 1.69 | 1.67 | 1.66 | .10      |                 |
| 2.34            | 2.27 | 2.19 | 2.15 | 2.11 | 2.06 | 2.02 | 1.97 | 1.94 | 1.92 | .05      |                 |
| 2.77            | 2.67 | 2.56 | 2.50 | 2.44 | 2.38 | 2.32 | 2.26 | 2.22 | 2.19 | .025     |                 |
| 3.37            | 3.23 | 3.08 | 3.00 | 2.92 | 2.84 | 2.75 | 2.66 | 2.61 | 2.57 | .01      |                 |
| 3.86            | 3.68 | 3.50 | 3.40 | 3.30 | 3.20 | 3.10 | 2.99 | 2.93 | 2.87 | .005     |                 |
| 5.13            | 4.87 | 4.59 | 4.45 | 4.30 | 4.15 | 4.00 | 3.84 | 3.75 | 3.67 | .001     |                 |

**TABLE 8**Percentage points of the  $F$  distribution ( $df_2$  between 19 and 24)

| $df_2$    | $\alpha$ | $df_1$ |       |      |      |      |      |      |      |      |      |
|-----------|----------|--------|-------|------|------|------|------|------|------|------|------|
|           |          | 1      | 2     | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
| <b>19</b> | .25      | 1.41   | 1.49  | 1.49 | 1.47 | 1.46 | 1.44 | 1.43 | 1.42 | 1.41 | 1.41 |
|           | .10      | 2.99   | 2.61  | 2.40 | 2.27 | 2.18 | 2.11 | 2.06 | 2.02 | 1.98 | 1.96 |
|           | .05      | 4.38   | 3.52  | 3.13 | 2.90 | 2.74 | 2.63 | 2.54 | 2.48 | 2.42 | 2.38 |
|           | .025     | 5.92   | 4.51  | 3.90 | 3.56 | 3.33 | 3.17 | 3.05 | 2.96 | 2.88 | 2.82 |
|           | .01      | 8.18   | 5.93  | 5.01 | 4.50 | 4.17 | 3.94 | 3.77 | 3.63 | 3.52 | 3.43 |
|           | .005     | 10.07  | 7.09  | 5.92 | 5.27 | 4.85 | 4.56 | 4.34 | 4.18 | 4.04 | 3.93 |
|           | .001     | 15.08  | 10.16 | 8.28 | 7.27 | 6.62 | 6.18 | 5.85 | 5.59 | 5.39 | 5.22 |
| <b>20</b> | .25      | 1.40   | 1.49  | 1.48 | 1.47 | 1.45 | 1.44 | 1.43 | 1.42 | 1.41 | 1.40 |
|           | .10      | 2.97   | 2.59  | 2.38 | 2.25 | 2.16 | 2.09 | 2.04 | 2.00 | 1.96 | 1.94 |
|           | .05      | 4.35   | 3.49  | 3.10 | 2.87 | 2.71 | 2.60 | 2.51 | 2.45 | 2.39 | 2.35 |
|           | .025     | 5.87   | 4.46  | 3.86 | 3.51 | 3.29 | 3.13 | 3.01 | 2.91 | 2.84 | 2.77 |
|           | .01      | 8.10   | 5.85  | 4.94 | 4.43 | 4.10 | 3.87 | 3.70 | 3.56 | 3.46 | 3.37 |
|           | .005     | 9.94   | 6.99  | 5.82 | 5.17 | 4.76 | 4.47 | 4.26 | 4.09 | 3.96 | 3.85 |
|           | .001     | 14.82  | 9.95  | 8.10 | 7.10 | 6.46 | 6.02 | 5.69 | 5.44 | 5.24 | 5.08 |
| <b>21</b> | .25      | 1.40   | 1.48  | 1.48 | 1.46 | 1.44 | 1.43 | 1.42 | 1.41 | 1.40 | 1.39 |
|           | .10      | 2.96   | 2.57  | 2.36 | 2.23 | 2.14 | 2.08 | 2.02 | 1.98 | 1.95 | 1.92 |
|           | .05      | 4.32   | 3.47  | 3.07 | 2.84 | 2.68 | 2.57 | 2.49 | 2.42 | 2.37 | 2.32 |
|           | .025     | 5.83   | 4.42  | 3.82 | 3.48 | 3.25 | 3.09 | 2.97 | 2.87 | 2.80 | 2.73 |
|           | .01      | 8.02   | 5.78  | 4.87 | 4.37 | 4.04 | 3.81 | 3.64 | 3.51 | 3.40 | 3.31 |
|           | .005     | 9.83   | 6.89  | 5.73 | 5.09 | 4.68 | 4.39 | 4.18 | 4.01 | 3.88 | 3.77 |
|           | .001     | 14.59  | 9.77  | 7.94 | 6.95 | 6.32 | 5.88 | 5.56 | 5.31 | 5.11 | 4.95 |
| <b>22</b> | .25      | 1.40   | 1.48  | 1.47 | 1.45 | 1.44 | 1.42 | 1.41 | 1.40 | 1.39 | 1.39 |
|           | .10      | 2.95   | 2.56  | 2.35 | 2.22 | 2.13 | 2.06 | 2.01 | 1.97 | 1.93 | 1.90 |
|           | .05      | 4.30   | 3.44  | 3.05 | 2.82 | 2.66 | 2.55 | 2.46 | 2.40 | 2.34 | 2.30 |
|           | .025     | 5.79   | 4.38  | 3.78 | 3.44 | 3.22 | 3.05 | 2.93 | 2.84 | 2.76 | 2.70 |
|           | .01      | 7.95   | 5.72  | 4.82 | 4.31 | 3.99 | 3.76 | 3.59 | 3.45 | 3.35 | 3.26 |
|           | .005     | 9.73   | 6.81  | 5.65 | 5.02 | 4.61 | 4.32 | 4.11 | 3.94 | 3.81 | 3.70 |
|           | .001     | 14.38  | 9.61  | 7.80 | 6.81 | 6.19 | 5.76 | 5.44 | 5.19 | 4.99 | 4.83 |
| <b>23</b> | .25      | 1.39   | 1.47  | 1.47 | 1.45 | 1.43 | 1.42 | 1.41 | 1.40 | 1.39 | 1.38 |
|           | .10      | 2.94   | 2.55  | 2.34 | 2.21 | 2.11 | 2.05 | 1.99 | 1.95 | 1.92 | 1.89 |
|           | .05      | 4.28   | 3.42  | 3.03 | 2.80 | 2.64 | 2.53 | 2.44 | 2.37 | 2.32 | 2.27 |
|           | .025     | 5.75   | 4.35  | 3.75 | 3.41 | 3.18 | 3.02 | 2.90 | 2.81 | 2.73 | 2.67 |
|           | .01      | 7.88   | 5.66  | 4.76 | 4.26 | 3.94 | 3.71 | 3.54 | 3.41 | 3.30 | 3.21 |
|           | .005     | 9.63   | 6.73  | 5.58 | 4.95 | 4.54 | 4.26 | 4.05 | 3.88 | 3.75 | 3.64 |
|           | .001     | 14.20  | 9.47  | 7.67 | 6.70 | 6.08 | 5.65 | 5.33 | 5.09 | 4.89 | 4.73 |
| <b>24</b> | .25      | 1.39   | 1.47  | 1.46 | 1.44 | 1.43 | 1.41 | 1.40 | 1.39 | 1.38 | 1.38 |
|           | .10      | 2.93   | 2.54  | 2.33 | 2.19 | 2.10 | 2.04 | 1.98 | 1.94 | 1.91 | 1.88 |
|           | .05      | 4.26   | 3.40  | 3.01 | 2.78 | 2.62 | 2.51 | 2.42 | 2.36 | 2.30 | 2.25 |
|           | .025     | 5.72   | 4.32  | 3.72 | 3.38 | 3.15 | 2.99 | 2.87 | 2.78 | 2.70 | 2.64 |
|           | .01      | 7.82   | 5.61  | 4.72 | 4.22 | 3.90 | 3.67 | 3.50 | 3.36 | 3.26 | 3.17 |
|           | .005     | 9.55   | 6.66  | 5.52 | 4.89 | 4.49 | 4.20 | 3.99 | 3.83 | 3.69 | 3.59 |
|           | .001     | 14.03  | 9.34  | 7.55 | 6.59 | 5.98 | 5.55 | 5.23 | 4.99 | 4.80 | 4.64 |